CD4 T cell subsets:

Different cells do different jobs…

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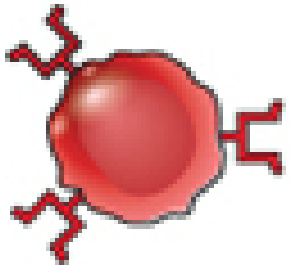
Aim of the session

- Part 1:
  - Brief overview CD4 T cells Subset Heterogeneity : Why? How? So what?
- Part 2:
  - Different subset: phenotype & functions
- Part 3
  - Heterogeneity, plasticity and flexibility: current view
  - Conclusion

T cell subsets



CD4 T cells

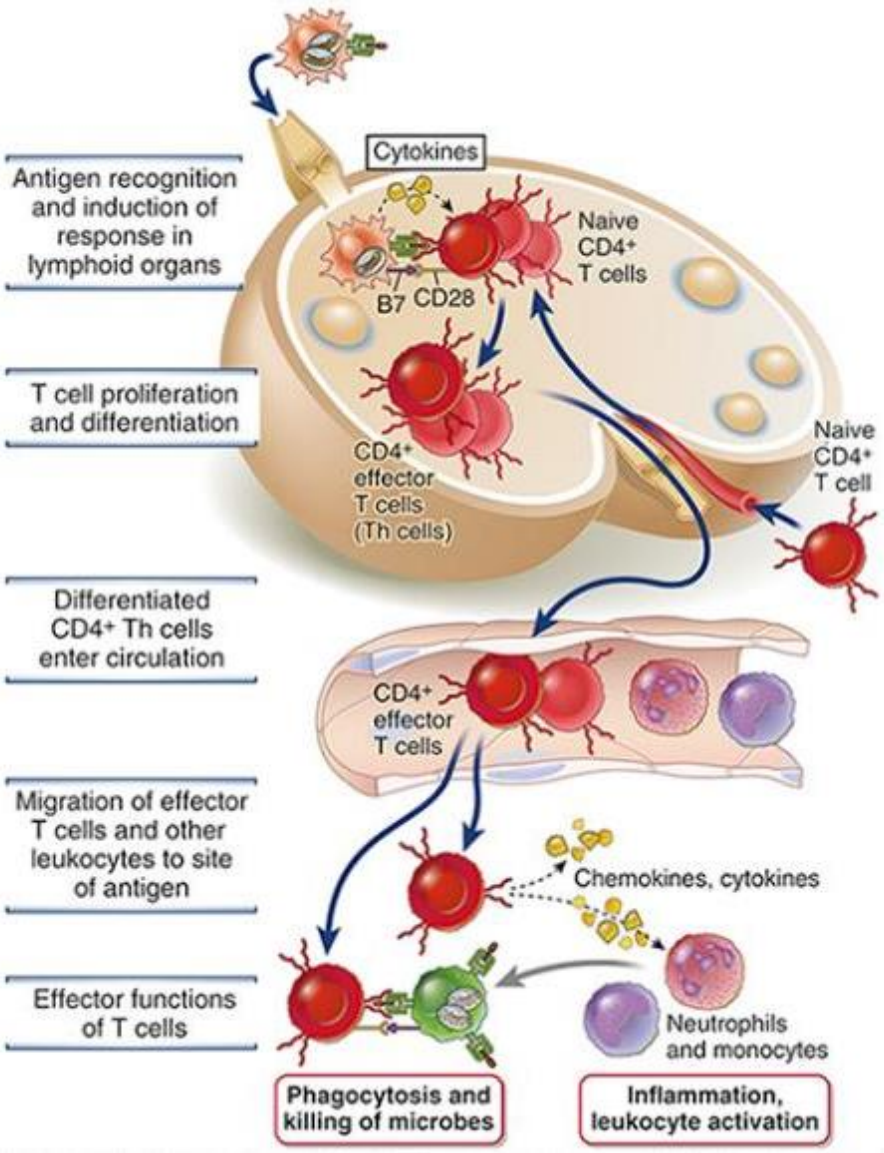
CD8 T cells

Cytotoxic T cells

Helper T cells
(Th cells)

Regulatory T cells
(suppress immune
responses)

Multitasking CD4 T cells

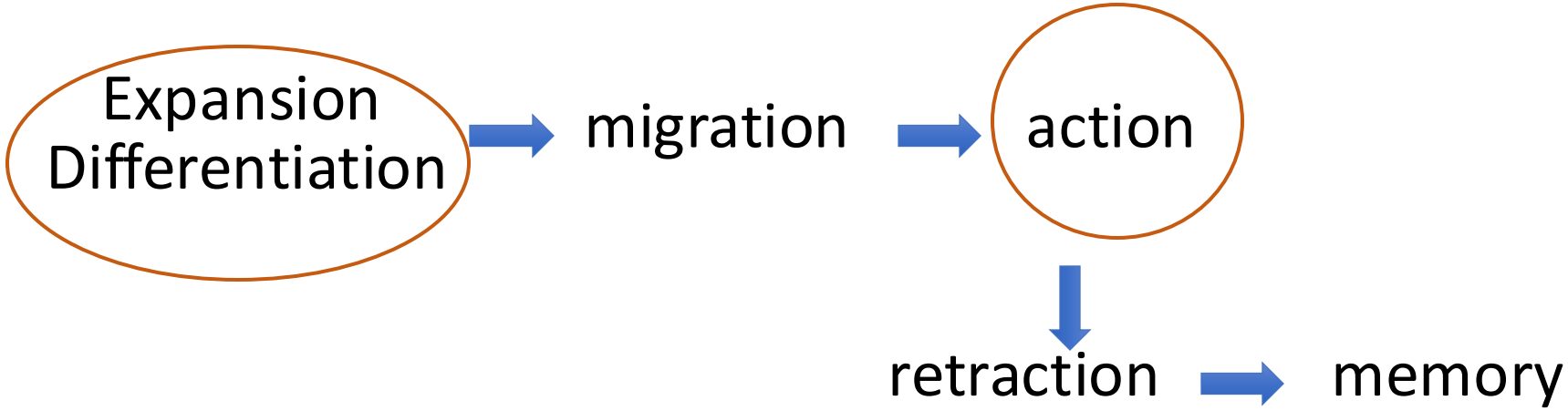


Protection of many different pathogens in different tissues:

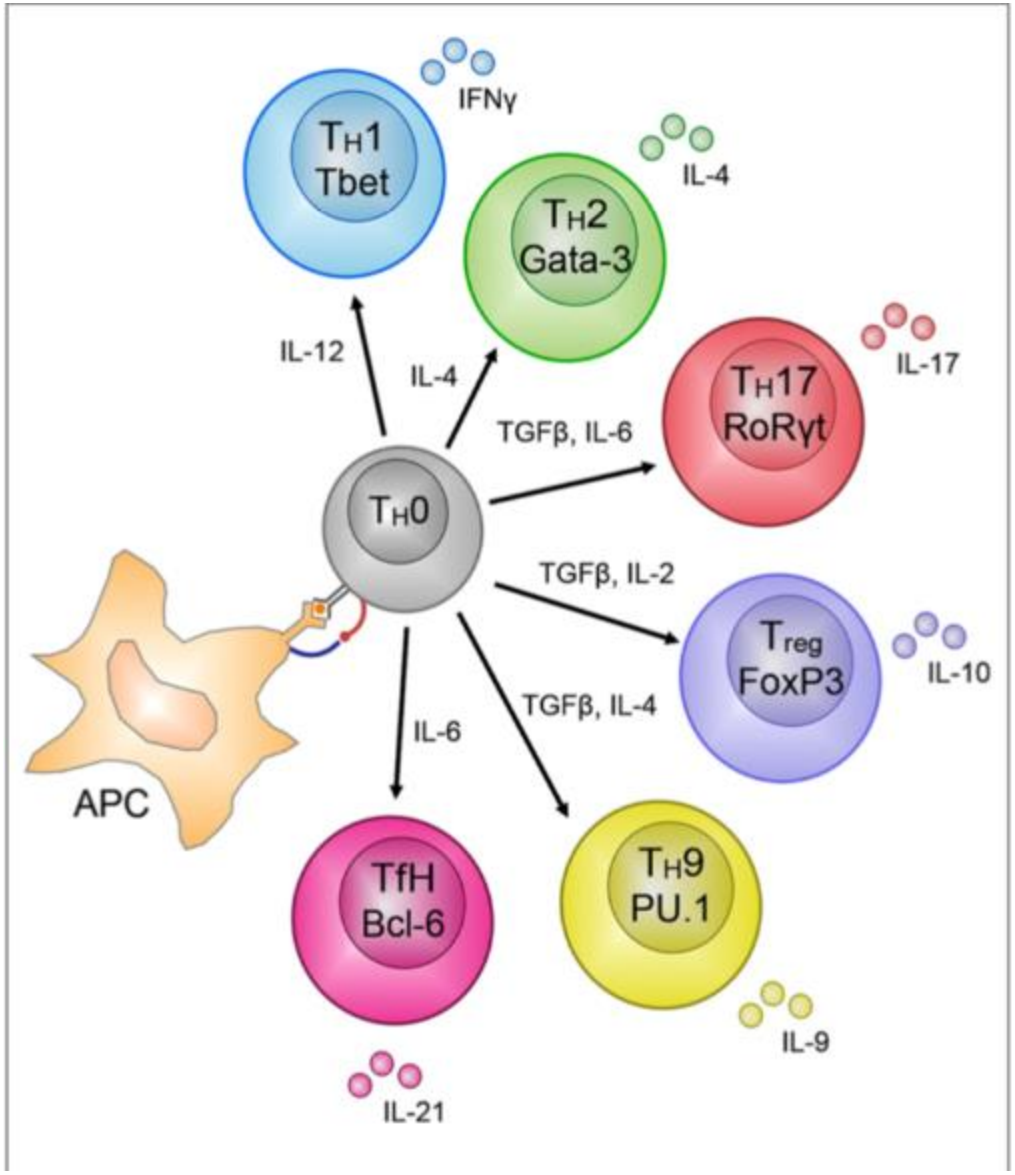
Many different types:

CD4 T cells become activated:

- Change in surface expression
- Change in migratory characteristics
- Change in functional capacity (differentiation)



CD4 Heterogeneity



- Why?
- How?
- So what?

APC deliver multiples signals to naïve CD4 T cells

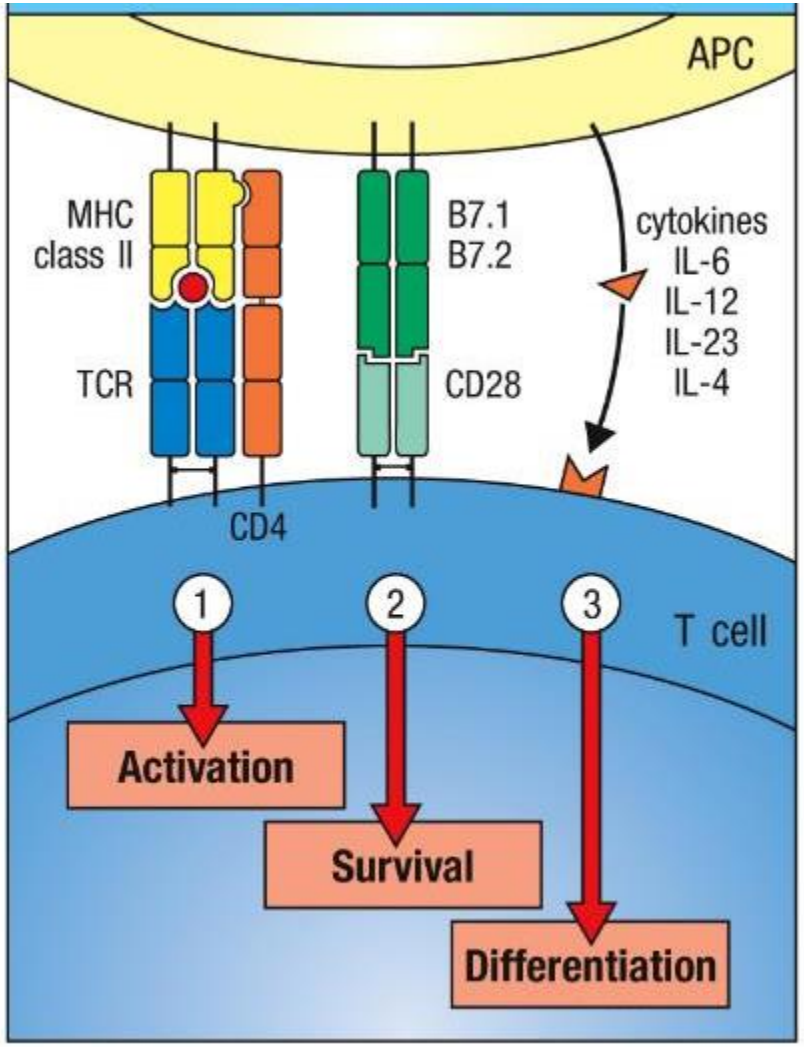


Figure 9.22 Janeway's Immunobiology, 9th ed. (© Garland Science 2017)

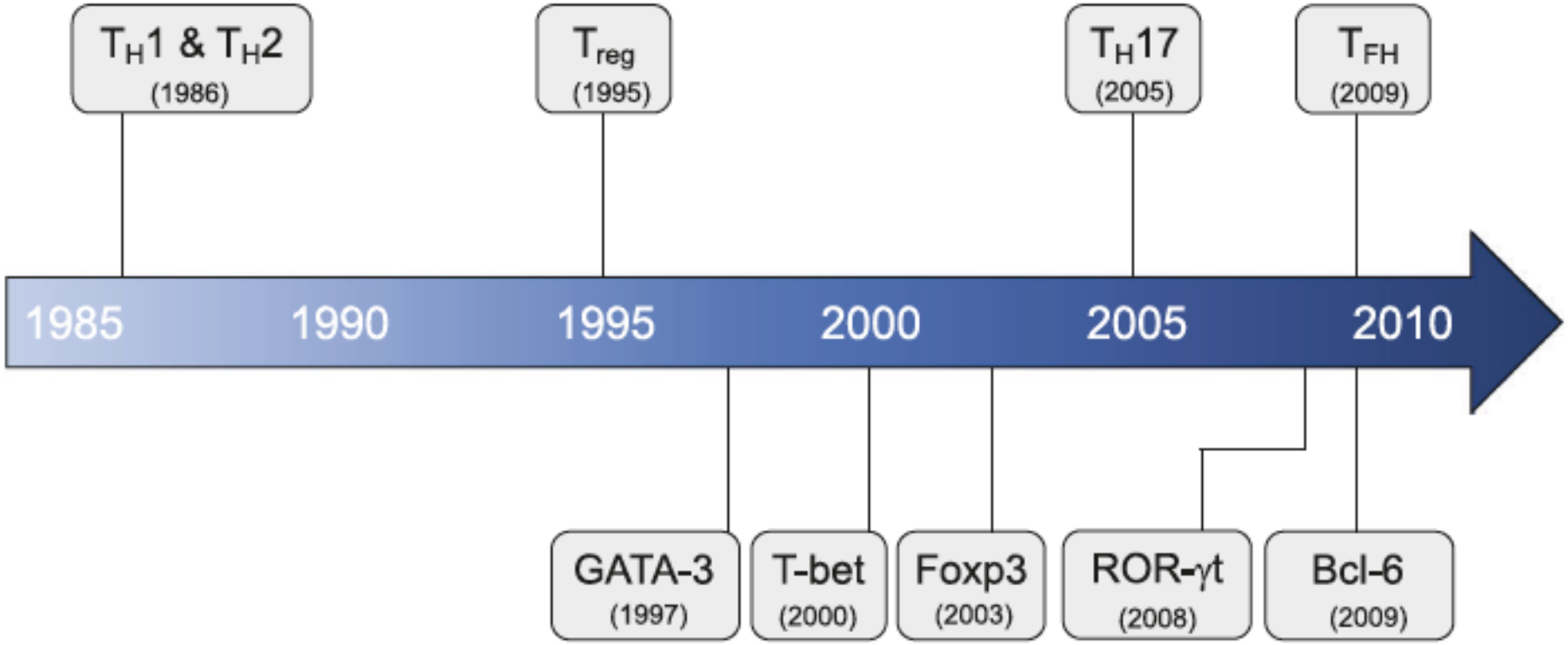
- Polarising cytokines
- Strength of TCR signal (dose, affinity, duration)
- Co-stimulation levels

Effector functions Th1 vs Th2

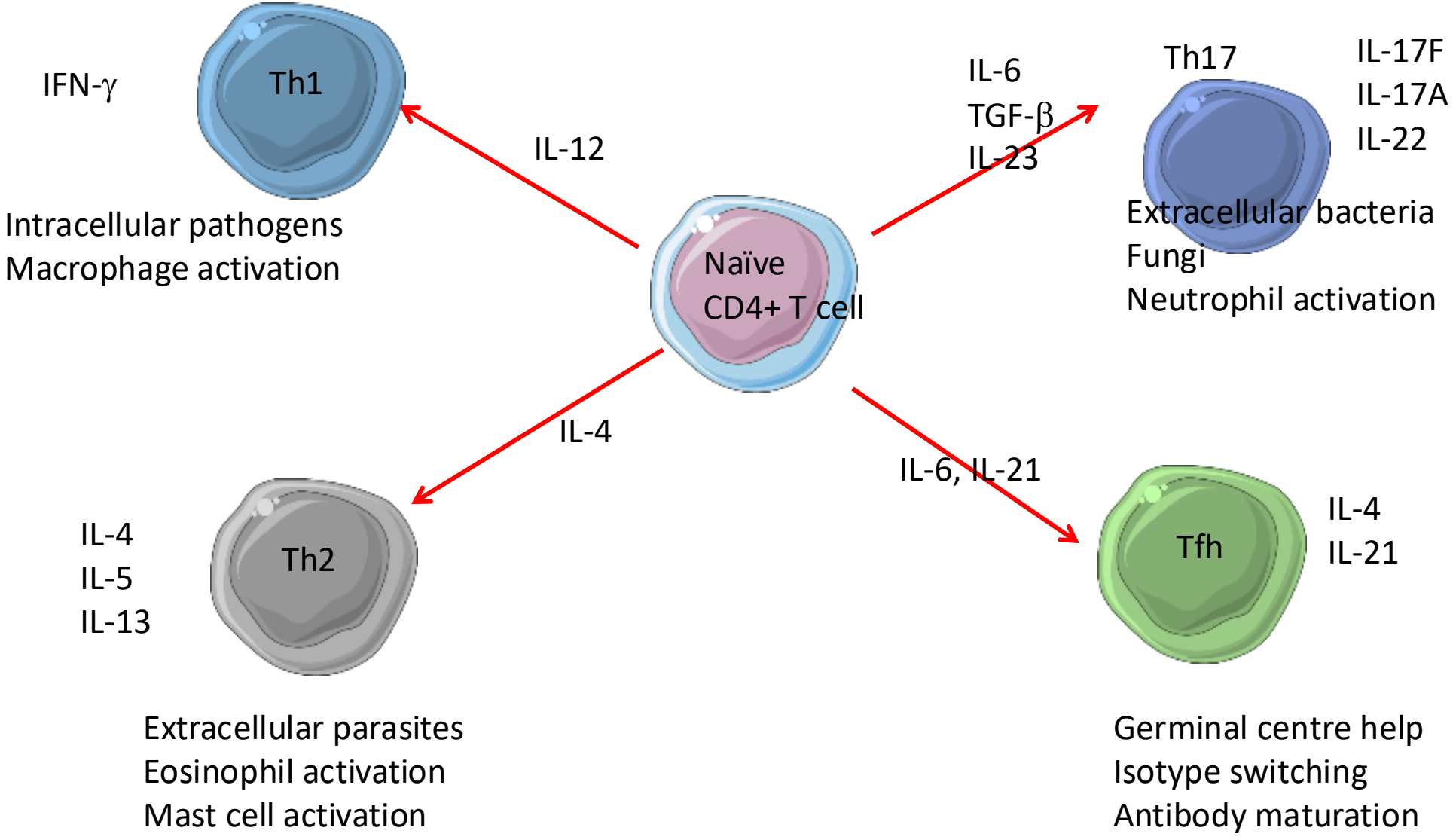
- Th1 vs Th2 response can be a major difference in the outcome
  - Leishmania (Balb c/ Th2 vs BL6 /Th1)
  - Human leishmania: localised skin disease or mucocutanous disease (Th1 vs mixed Th1/Th2)
  - Tuberculoid (Th1) vs lepromatous (Th2) leprosy

- IL-4 and IFNγ

More subsets…



CD4 Helper functions



Cytokines and transcription factors determine CD4 polarisation

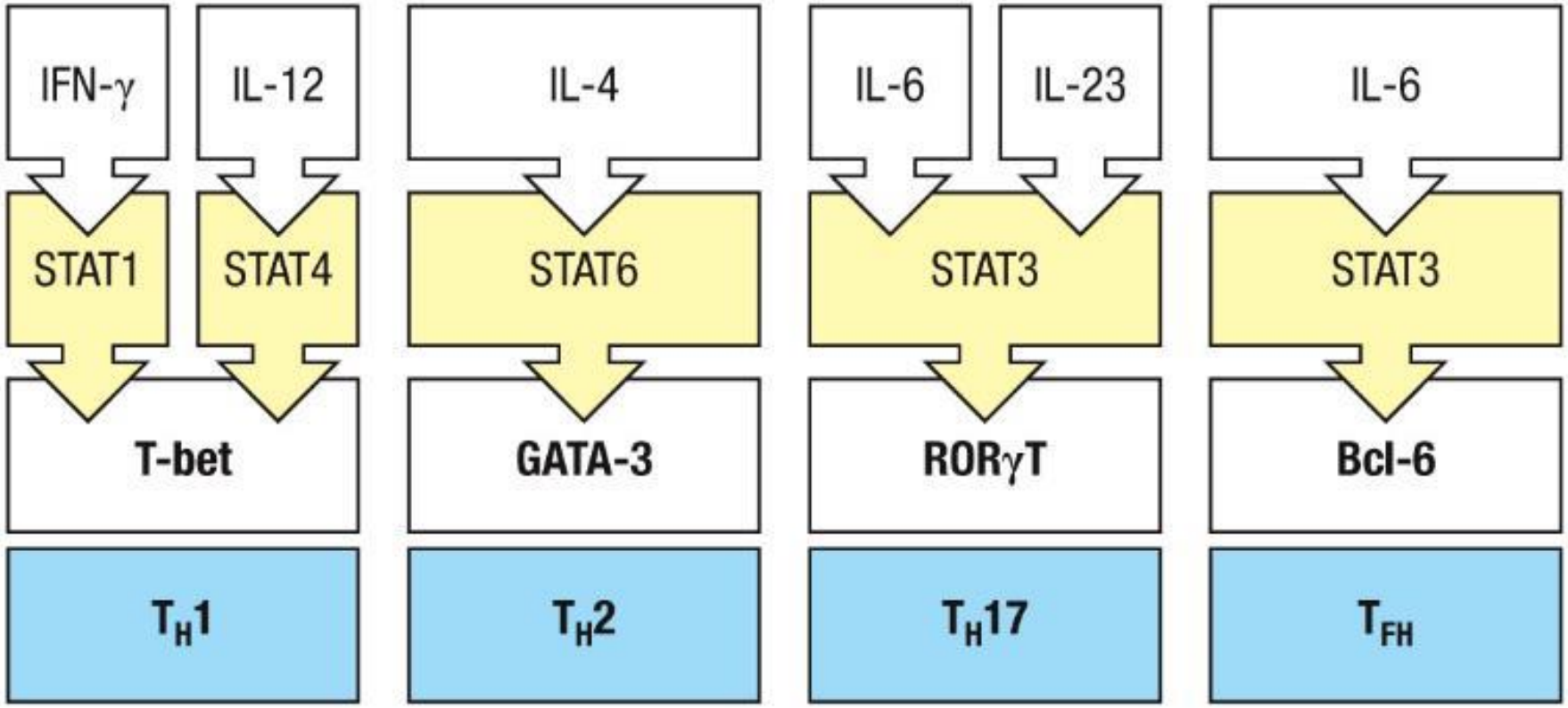


Figure 9.32 Janeway's Immunobiology, 9th ed. (© Garland Science 2017)

CD4 Polarisation

- Polarisation factors:
  - Antigen dose
  - Cytokine milieu
  - TCR – duration of signal
  - TCR- quality/ strength of signal
  - Co-stim level
  - Type of APC

External factors →

different signaling pathways

(STAT, NFkb, NFAT, mTOR) →

Signal transfer to the nucleus

- Chromatin remodelling
- Regulatory elements configured
- Recruit TF

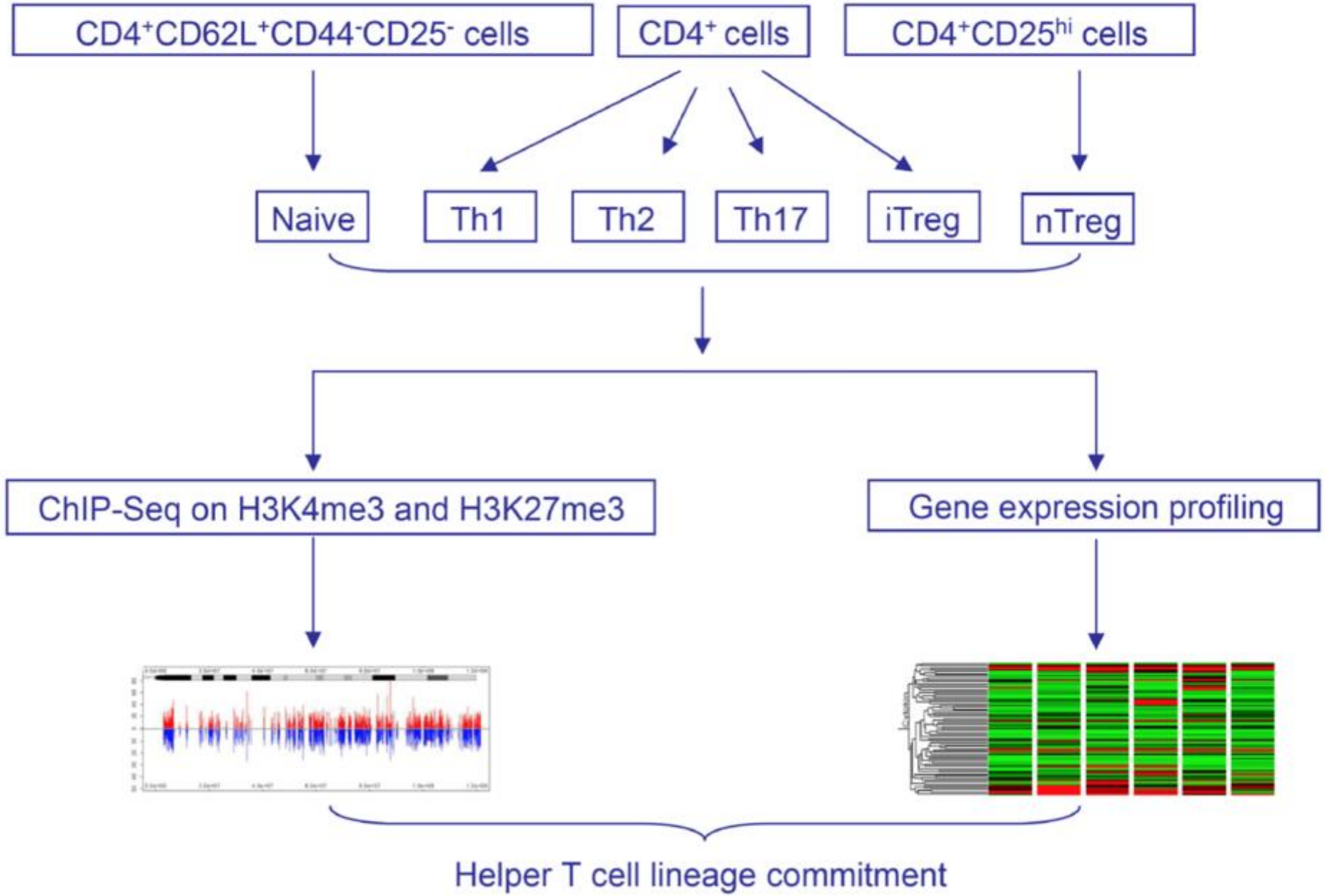
Lineage specific gene expression profiles

Lineage commitment

- Linage commitment: Cytokines activate STAT proteins (JAK-STAT pathway)
- ‘Master’ transcription factors (Tbet, GATA3, Rorγt): lineage specific
  - Mutually exclusive
  - Inhibit each other transcriptionally or protein interactions
  - One TF = 1 cell fate??
- What happens if the environment changes?

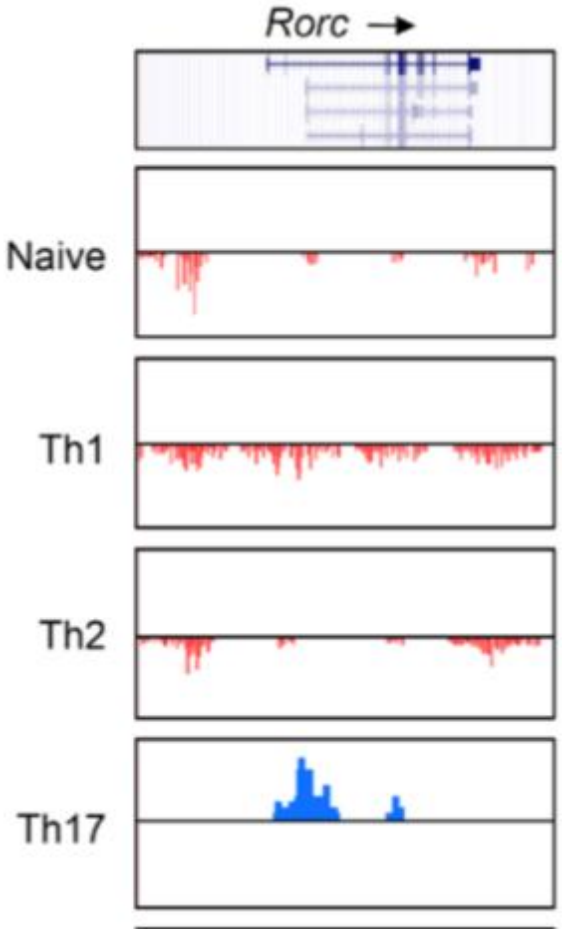
Master regulators and beyond....

- Master regulators: are TF factors enough to confer phenotype?
- Regulatory network-pioneering transcriptional factors (IRF4, STAT3, BATF for Th17)
- Epigenetic changes: chromatin accessibility: DNA methylation or acetylation



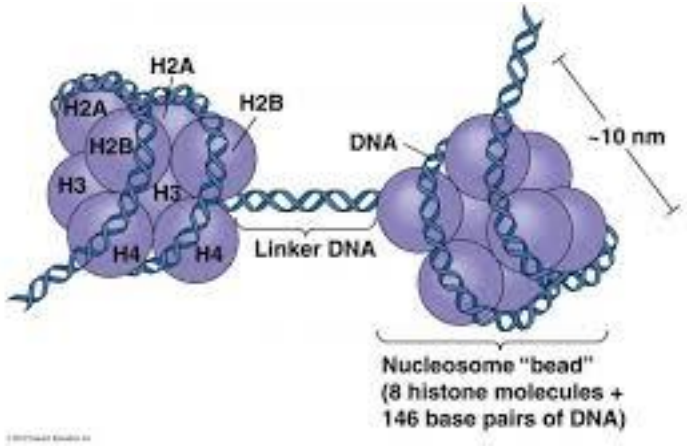
Wei G et al, Immunity 2009

Chromatin accessibility and master regulators



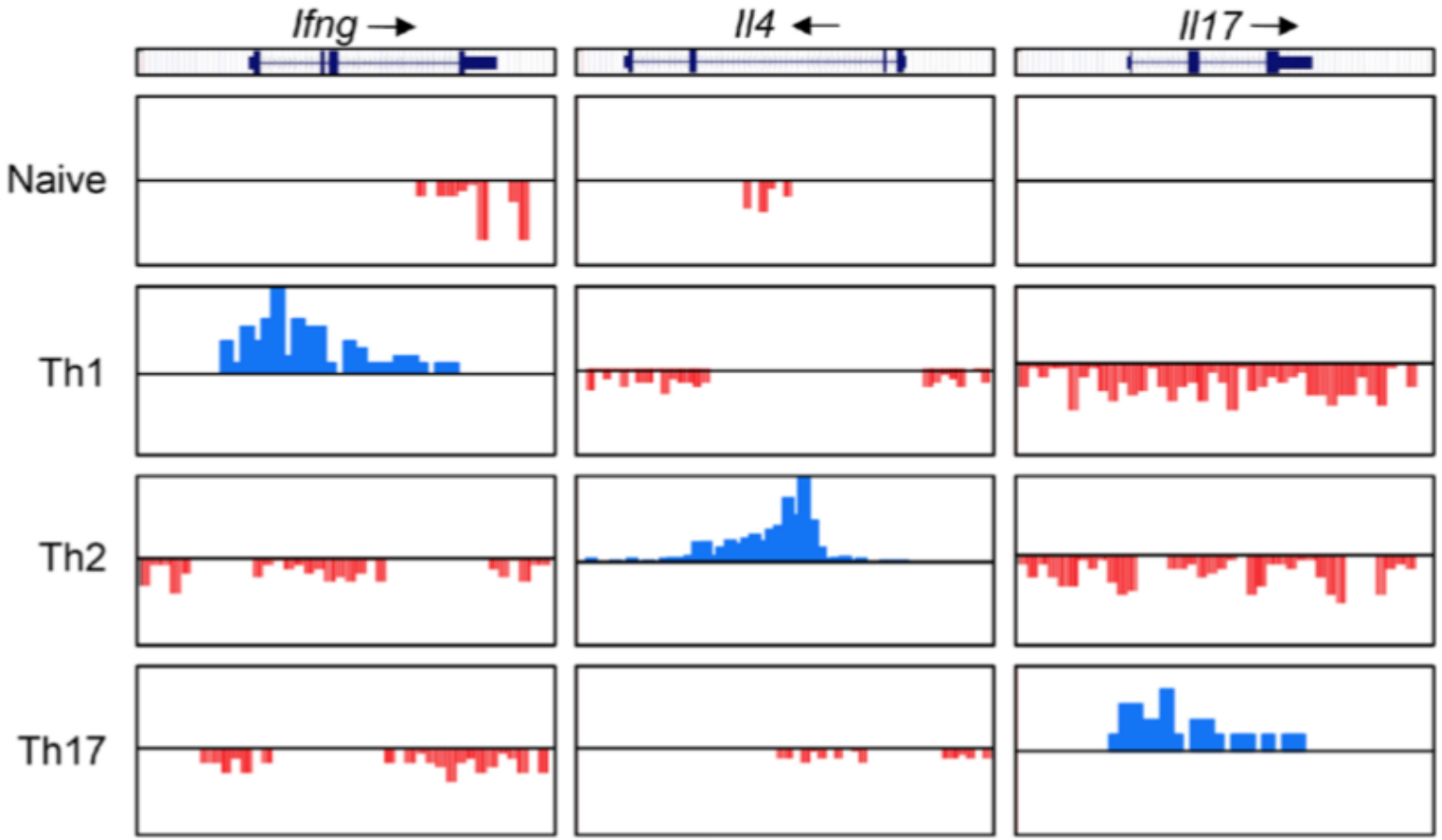
H3K27me3 : inactive genes

H3K4me3: active genes (promoters/ enhancers)



Wei G et al, Immunity 2009

Signature cytokines and gene accessibility



H3K27me3

H3K4me3

Wei G et al, Immunity 2009

Conclusions?

CD4 T cell subsets:

Part 2

Function of different subsets

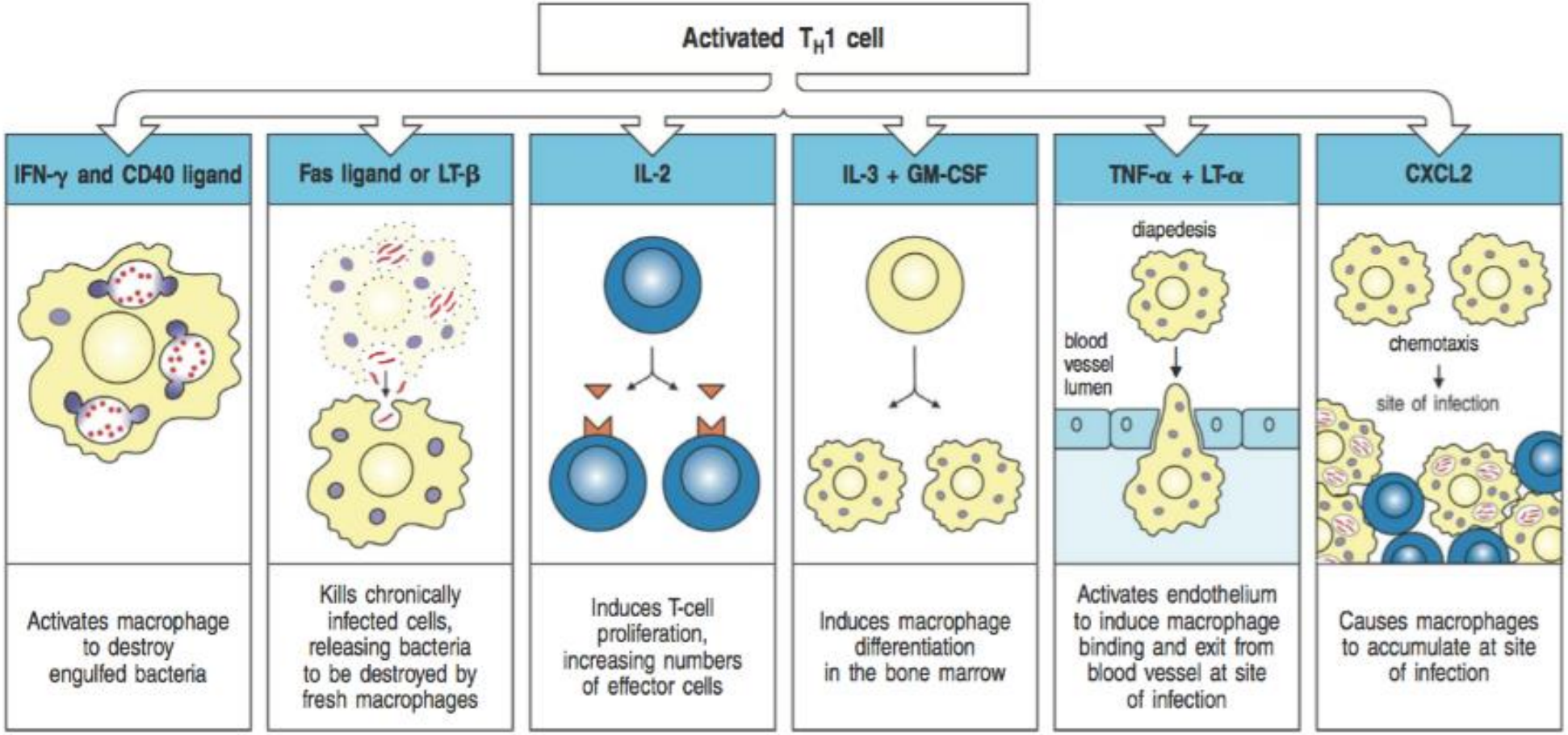
Th1 cells

- Th1 cells: IFN-γ, CXCR3, **T-bet**
- Immune responses to intracellular bacteria/ viruses
- Classic example Mycobacterium Tuberculosis: super-activation of macrophages

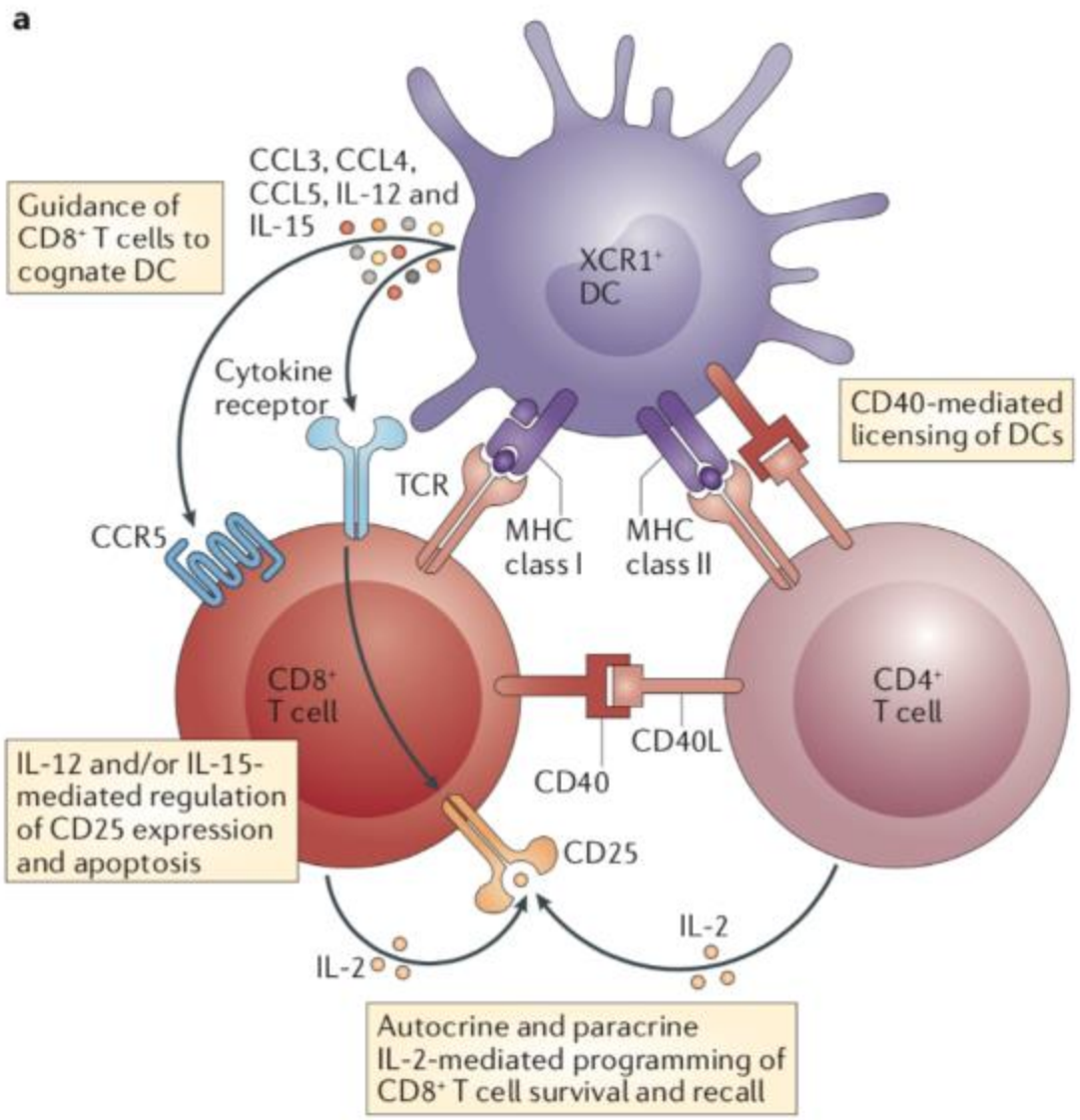
HOWEVER: Th1 cells in MTb infection express CCR6, CD161 and RORγt:

Th1* subset (transcriptionally different to classic Th1)

Th1: Macrophage help



Th1 cells: CD8 help



CD4+ T cell calibration of antigen-presenting cells optimizes antiviral CD8+ T cell immunity Gressier et al (2023): Nat Immunol:24:979–990

https://doi.org/10.1038/s41590-023-01517-x

Th1 cells in viral infections

- Produce TNF-α and IFN-γ
- Help for CTLs which kill infected cells
- IFNγ enhances IgG2a class switching (usually the dominant isotype in anti-viral antibody responses)
- Can acquire capacity to kill (FASL, perforin, Granzyme B)

Th2 cells

- Extracellular parasites, venoms, allergens
- IL-4, IL-13, IL-5
- STAT6, **GATA3**
- CCR4, CCR3, CrTh2
- Recruitment and activation of eosinophils, mast cells (also CCR3+)
- IL-5+ Th2: highly differentiated Th2 cells
- Can produce IL-9, and can convert into Th9 (TGFβ)

Th17

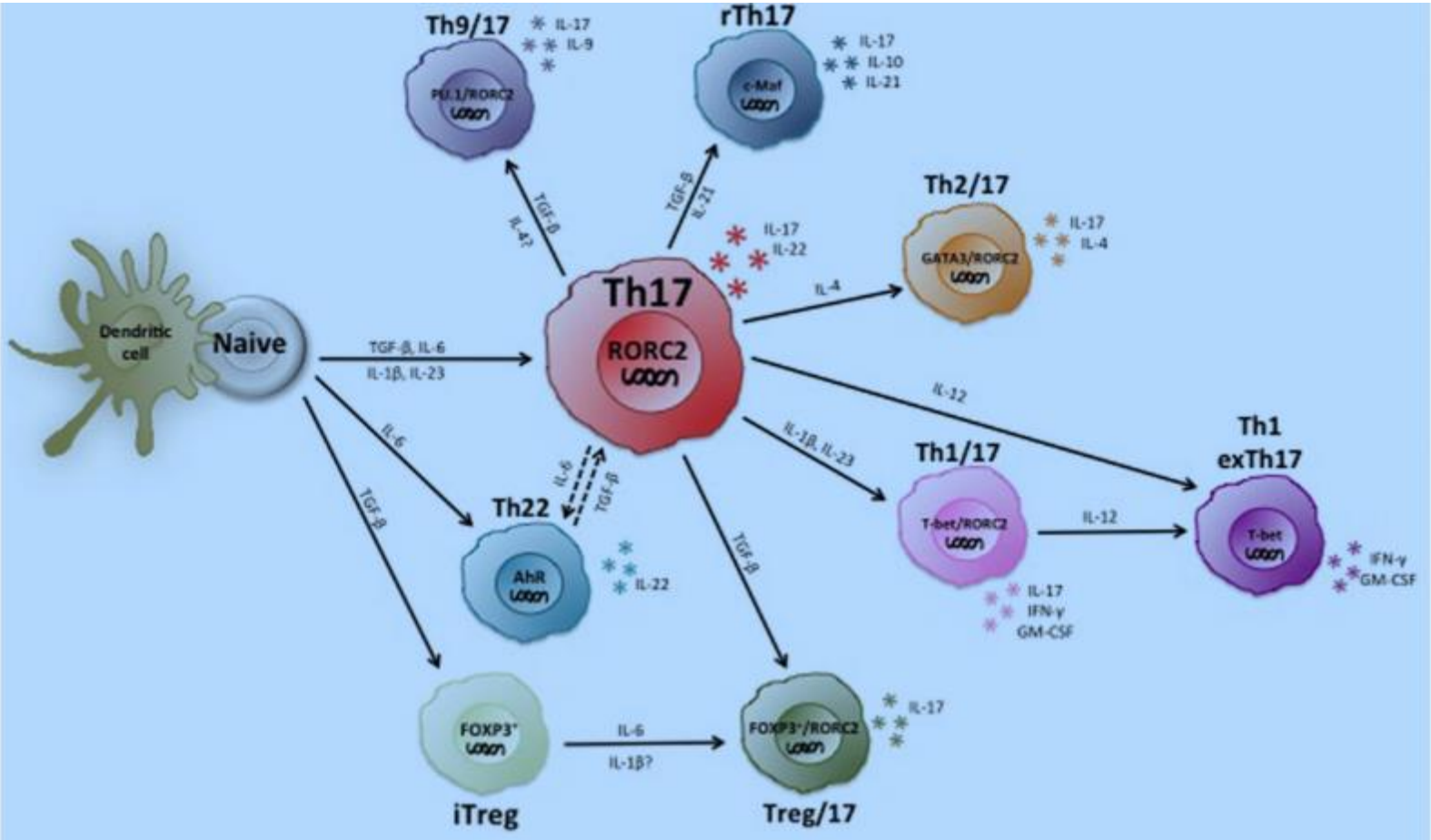
- Evidence from mouse autoimmunity models
- Protection against fungi and extracellular bacteria
- Enriched in barrier tissues
- Induced by TGF-β, IL-6 (in mice) BUT IL-6, TGF-β, IL-1β, and/or IL-23 (in humans)
- STAT3, **RORγt**
- CCR6

Th17- heterogeneity and instability

- Sometimes express IFNγ (C.albicans) or IL-10 (S.aureus) together with IL-17
- Conventional, regulatory, pathogenic Th17
- May not be as stable as other subsets, converting to Th1-like, Treg-like
- Often also produce IL-22, IL-21, IL-9
- Closely related to IL-22 secreting cells (CCR6, CLA, CCR10)

Th17 cell heterogeneity and its role in tissue inflammation
Schnell et al (2023):Nat Immunol 24:19–29

Th17

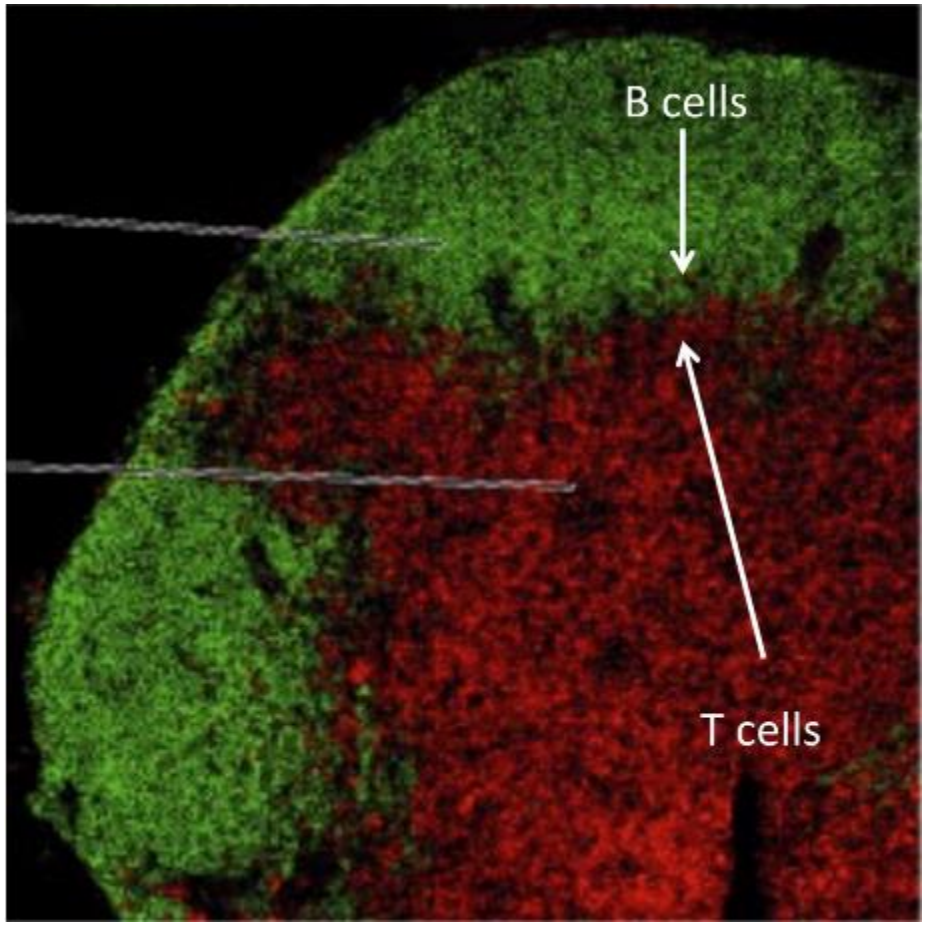

Th9/17
IL-17
IL-9
PU.1/RORC2
rTh17
IL-17
IL-10
IL-21
c-Maf
TGF-β
IL-4?
TGF-β
IL-21
Th2/17
IL-17
IL-4
GATA3/RORC2
IL-17
IL-22
Th17
RORC2
IL-4
Dendritic cell
Naive
TGF-β, IL-6
IL-1β, IL-23
IL-12
IL-6
IL-1β, IL-23
Th1
exTh17
Th1/17
IL-6
TGF-β
Th22
AhR
IL-22
T-bet/RORC2
IL-12
T-bet
IFN-γ
GM-CSF
IL-17
IFN-γ
GM-CSF
TGF-β
TGF-β
FOXP3+
IL-6
IL-1β?
FOXP3+/RORC2
IL-17
iTreg
Treg/17

Tfh –T Follicular helper cells

- IL-6 induced **Bcl-6,**
- Upon activation up-regulate CXCR5, migrate to B cell follicle (in response to CXCL13)
- Resident in germinal center, sustained interaction with B cells: IL-21, ICOS
- Specialised providers of help for B cells (germinal center formation, affinity maturation, high-affinity antibodies and memory B cell development)

LN: T cells help B cells

6. Germinal centers are formed within the B cell follicle. High affinity antibody is produced.



1.DC activate T cells (in the T cell zone)

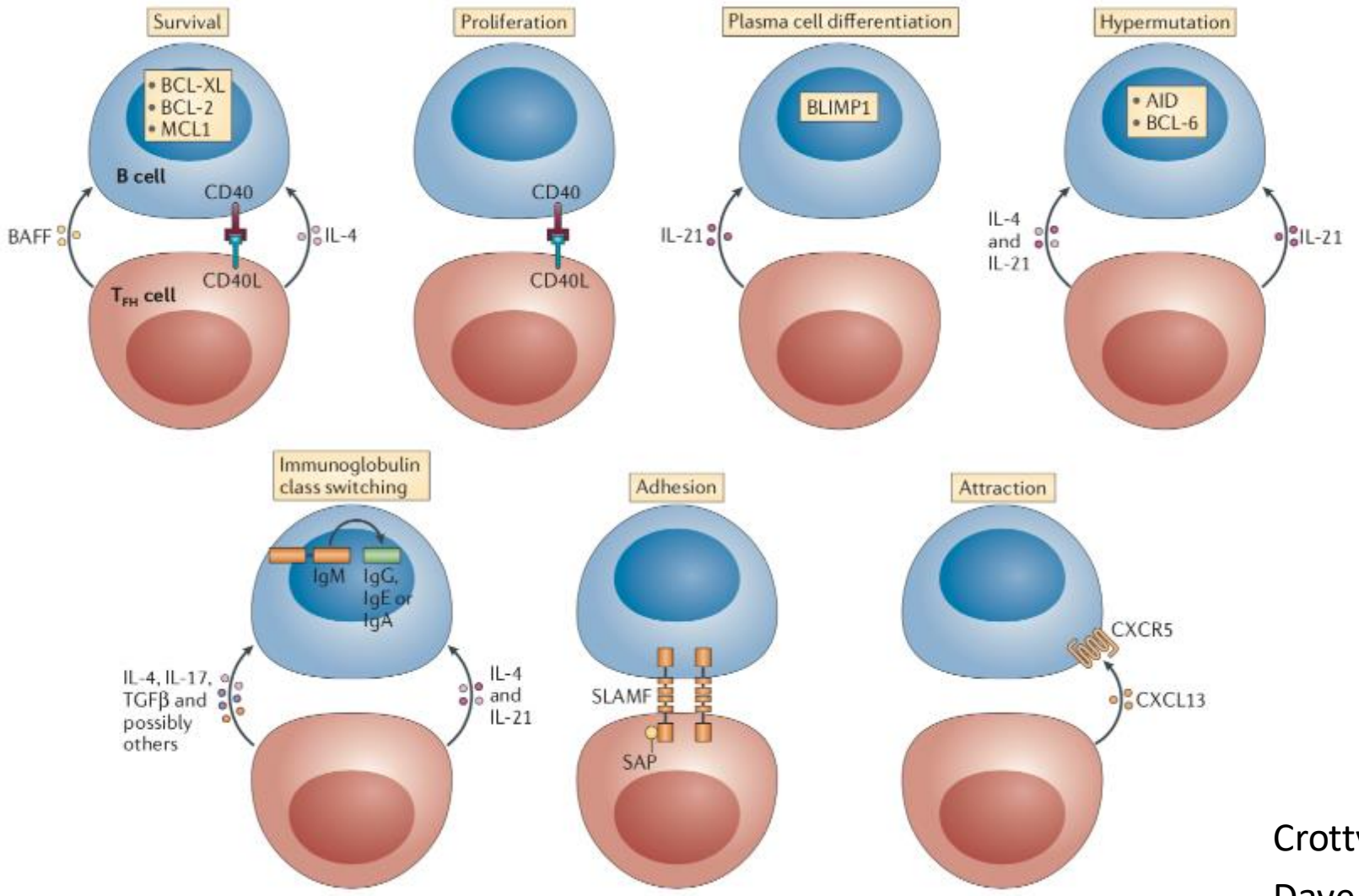
2. Primed CXCR5+ T cells move to T/B border

3. B cells that encounter Ag move to T/B border

4. T/B border: major site of T/ B cell interaction

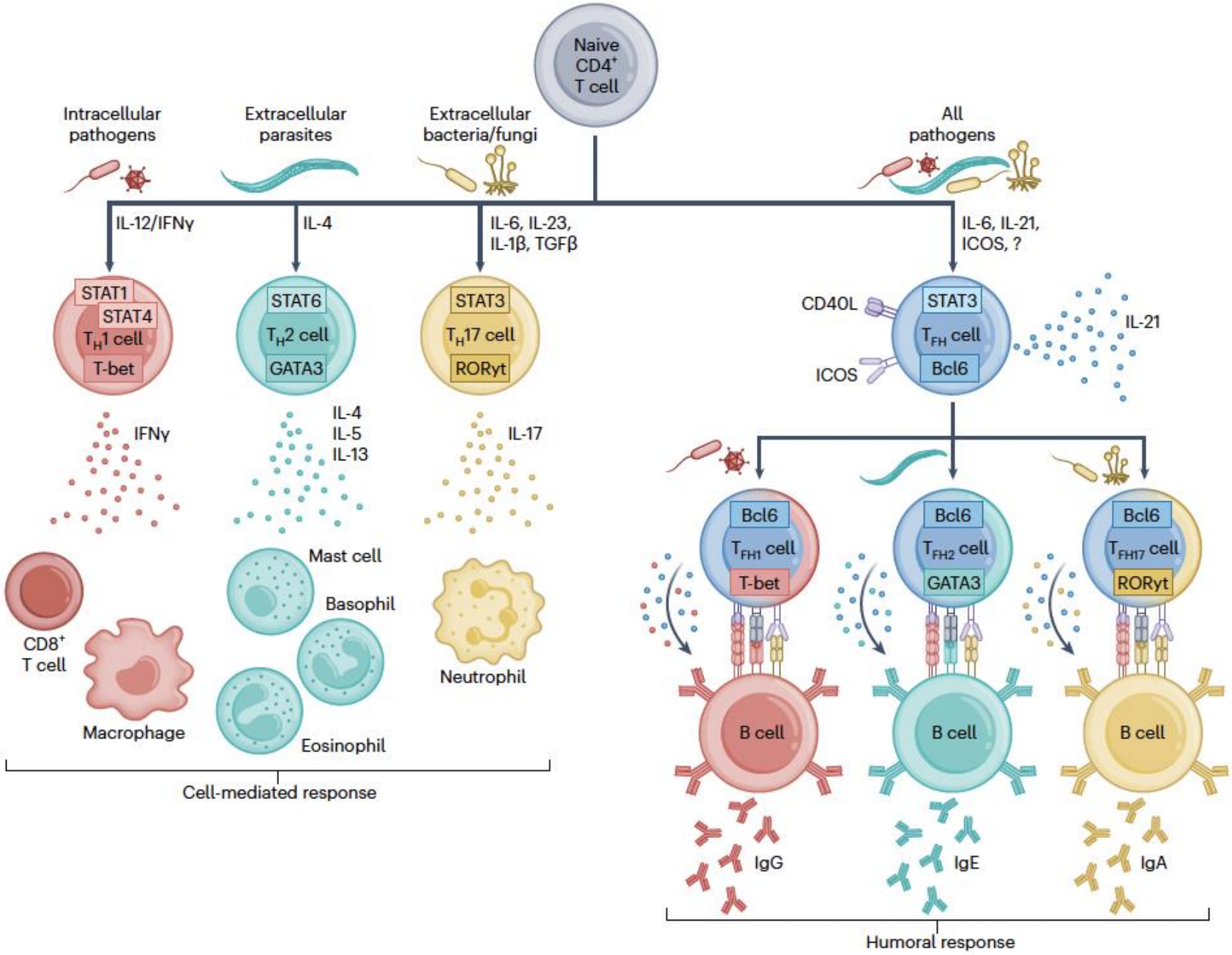
5. B cells may initiate an extrafollicular response (outside the B cell follicle) or a germinal center formation

Tfh- functions



Crotty Immunity, 2019

Dave & Ballesteros-Tato
Sci Immunol 2025


Naive CD4+ T cell
Intracellular pathogens
Extracellular parasites
Extracellular bacteria/fungi
All pathogens
IL-12/IFNγ
IL-4
IL-6, IL-23, IL-1β, TGFβ
IL-6, IL-21, ICOS, ?
STAT1
STAT4
TH1 cell
T-bet
STAT6
TH2 cell
GATA3
STAT3
TH17 cell
RORγt
CD40L
ICOS
STAT3
TFH cell
Bcl6
IL-21
IFNγ
IL-4
IL-5
IL-13
IL-17
Mast cell
Basophil
CD8+ T cell
Macrophage
Eosinophil
Neutrophil
Cell-mediated response
Bcl6
TFH1 cell
T-bet
Bcl6
TFH2 cell
GATA3
Bcl6
TFH17 cell
RORγt
B cell
B cell
B cell
IgG
IgE
IgA
Humoral response

CD4 T cell subsets:

Part 3

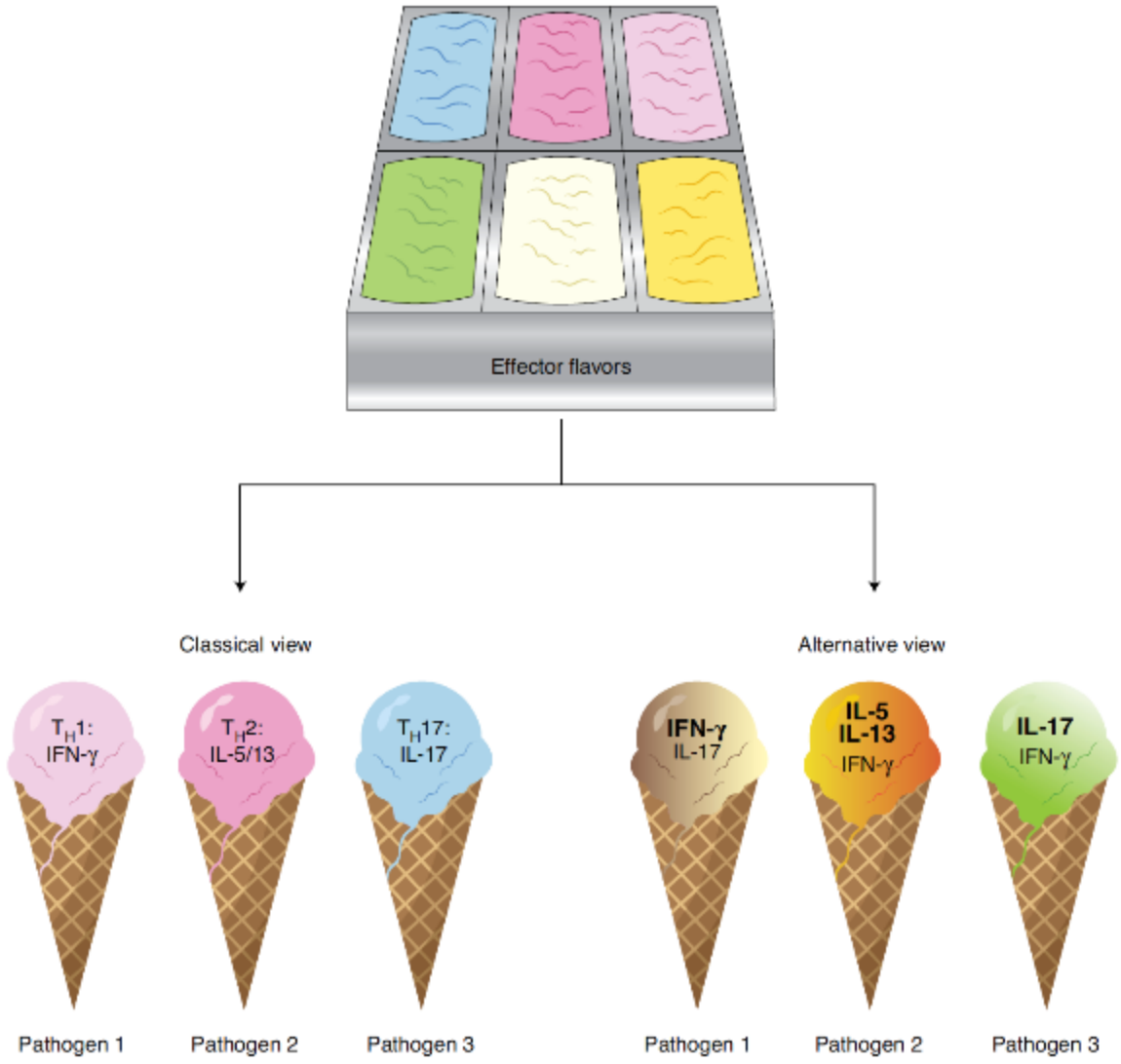
Subset heterogeneity and plasticity

Sallusto, Annu. Rev. Immunol. 2016. 34:317–34

Zemmour et al, Curr Opin Immunol. 2020. 63:61-67

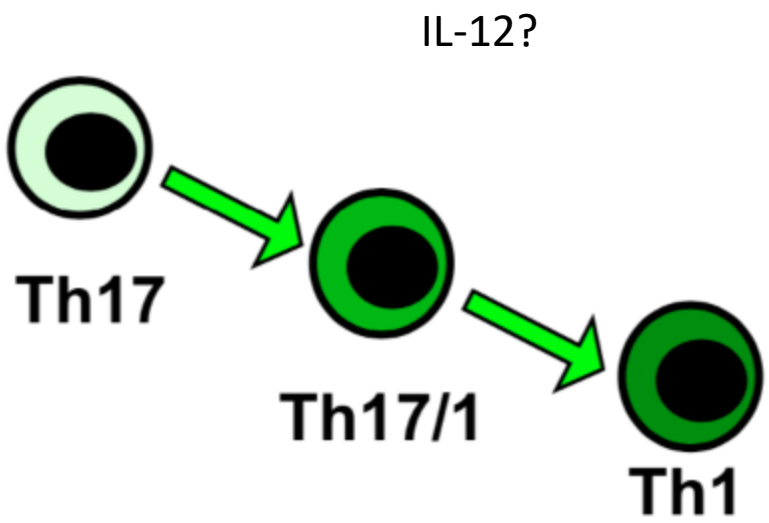
Tuzlak et al. Nat Immunol 2021. 22: 1210–1217

A fresh look at the T helper subset dogma



Subset plasticity:
How fixed is ‘differentiated state’

Lineage tracing experiments: Polarised subsets can alter phenotype during their lifespan



Th17 cells which switch to ‘Th1’ cells are still CD161+ve, CCR6+ve

New view on subset heterogeneity

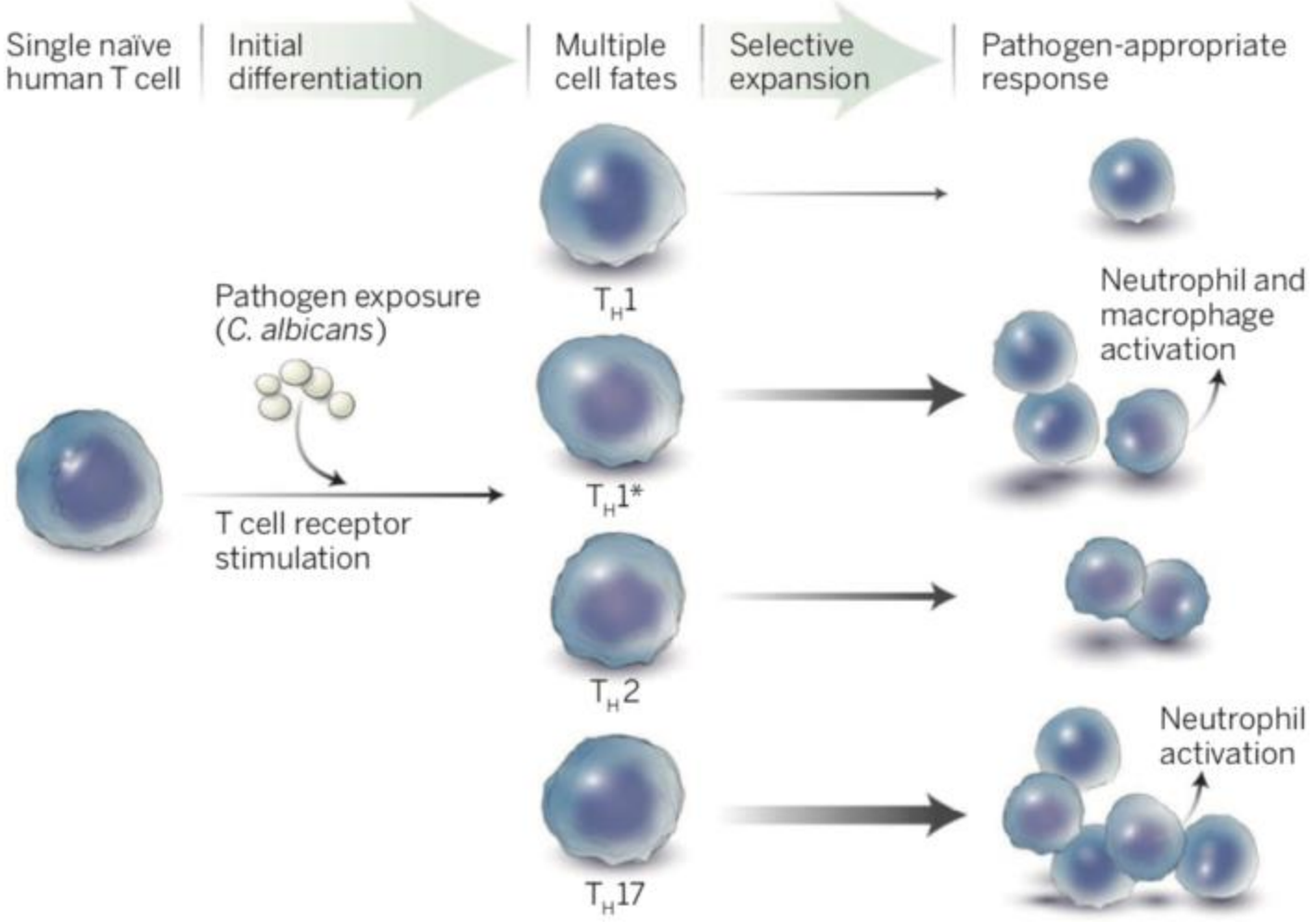
- The classic view: different antigen need different type of response
- Evidence that within a pool of antigen specific cells the majority may be one type (Th1, Th17) BUT others could be of a different type
- Complexity of antigens/ many different PAMP

- Extremely heterogeneous antigen specific populations (interclonal heterogeneity?)

Becattini S et al *Science* 347:400–6

Different phenotypes: Cell subsets (and sub-subsets) or cell states?

Flexibility: One cell multiple fates?

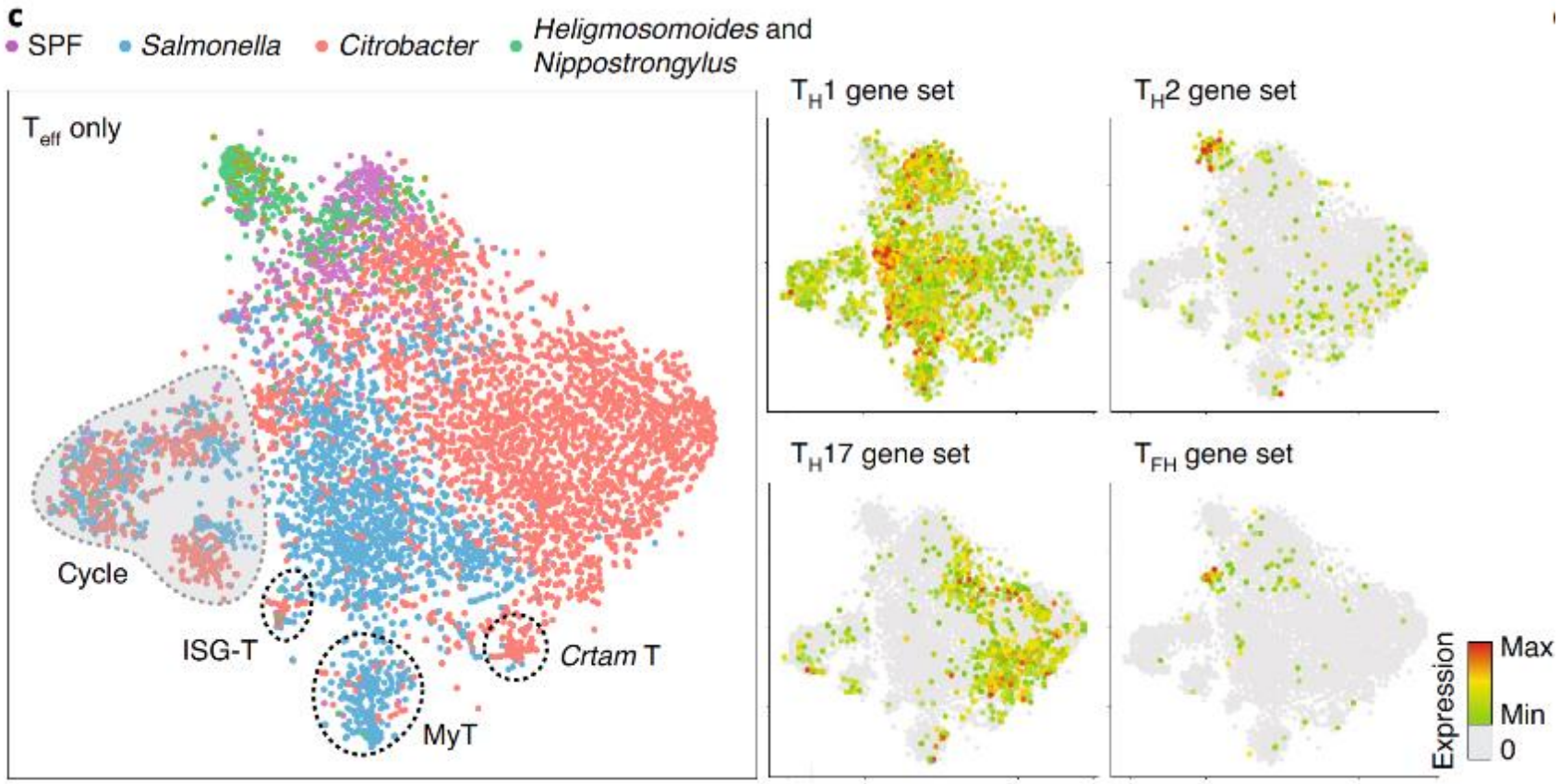


1. Different types of Teff for a single pathogen

2. Single naïve T cell can adopt different fate/ phenotype

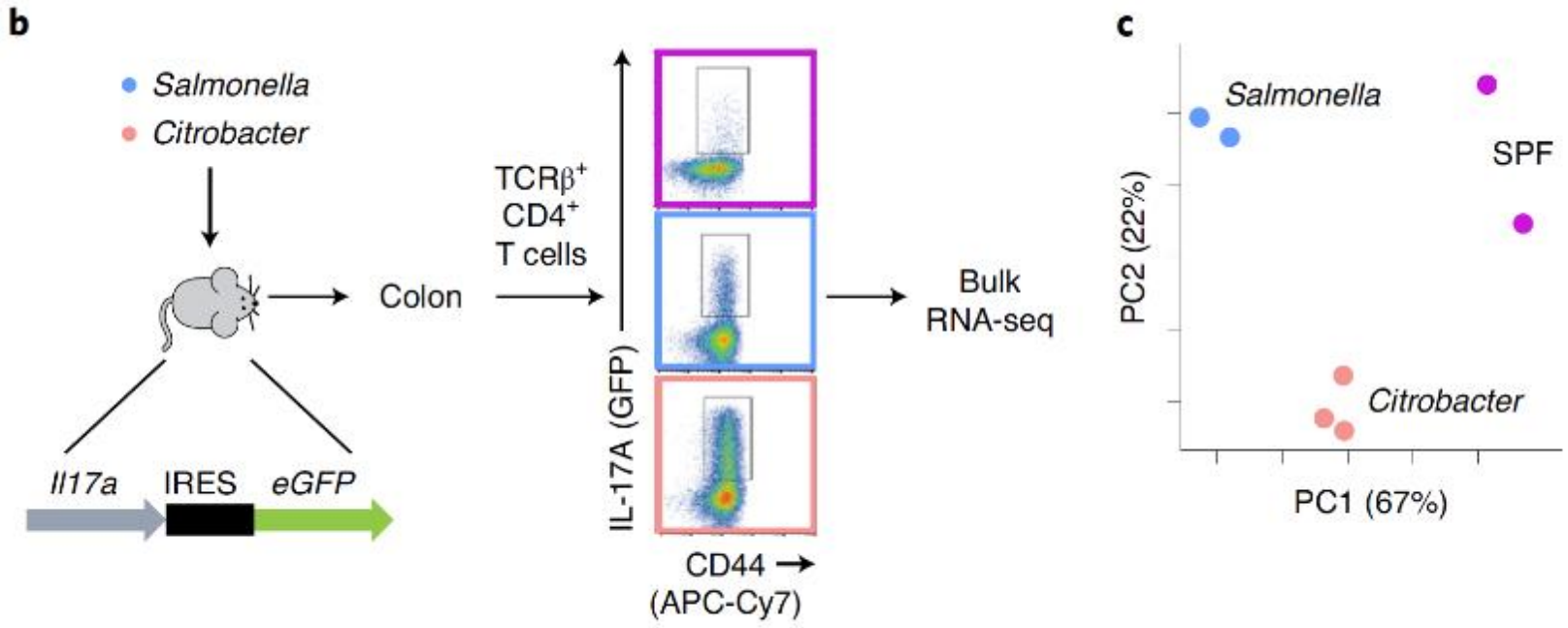
Becattini S et al *Science* 347:400–6
F. Sallusto. *Annu Rev Immunol* 2016. 34:317–34

Lack of discrete subsets using single-cell-sequencing approaches



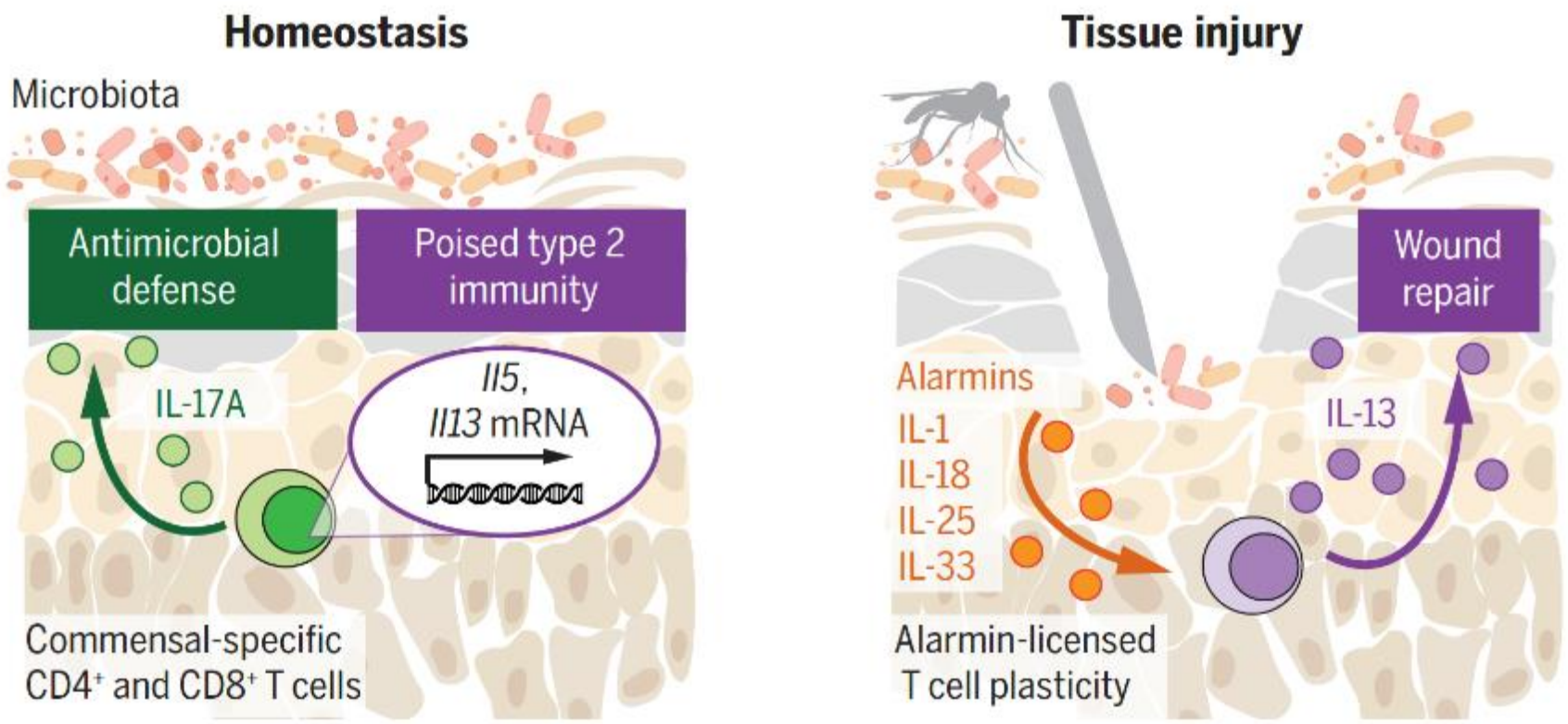
Kiner et al, (2021) Nature Immunology 22, 666–668

Pathogen driven response in the gut



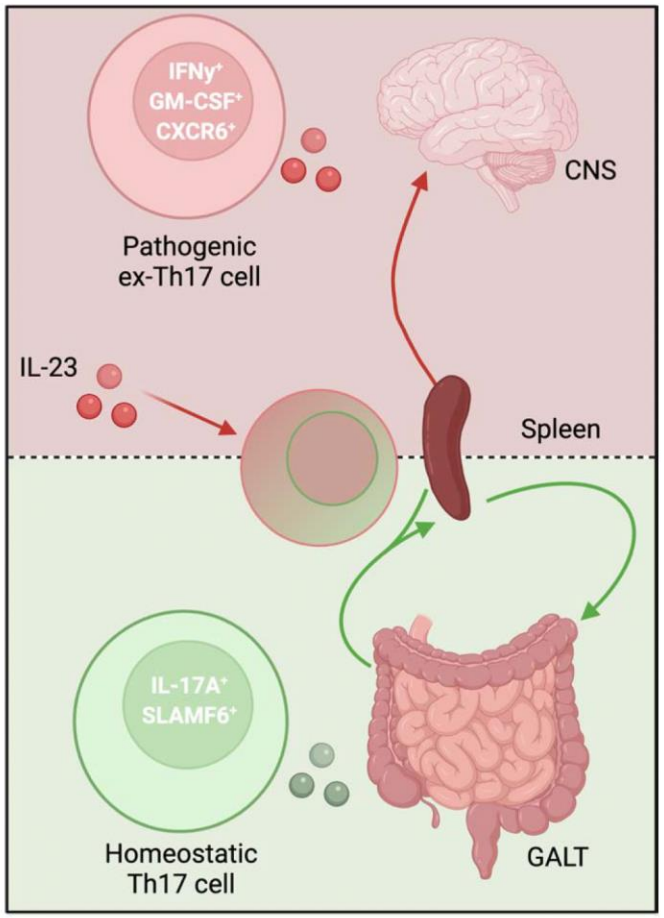
Kiner et al, (2021) Nature Immunology 22, 666–668

Microbiota specific T cell plasticity in the skin



Harrison et al., Science 363, 43 (2019)

Intestinal Th17 cells give rise to pathogenic effector T cells during autoimmunity



Combined scRNA/TCR-seq reveals extensive heterogeneity of tissue Th17 cells

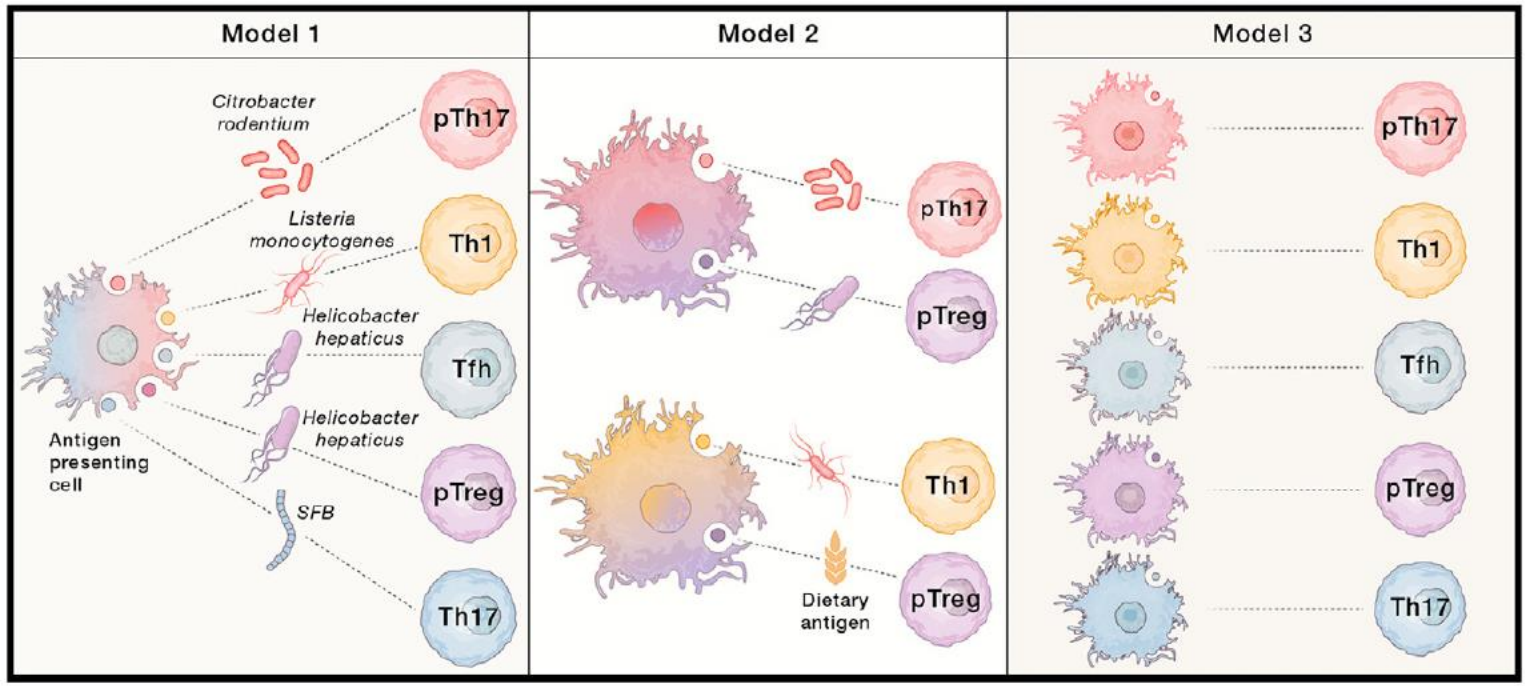
Th17 cells exhibit strong tissue-specific signatures

Pathogenic CXCR6+ Th17 populations are induced in autoimmunity

IL-23 converts the stem-like/ homeostatic IL-17+ to the pathogenic GM-CSF + IFN-γ+ population

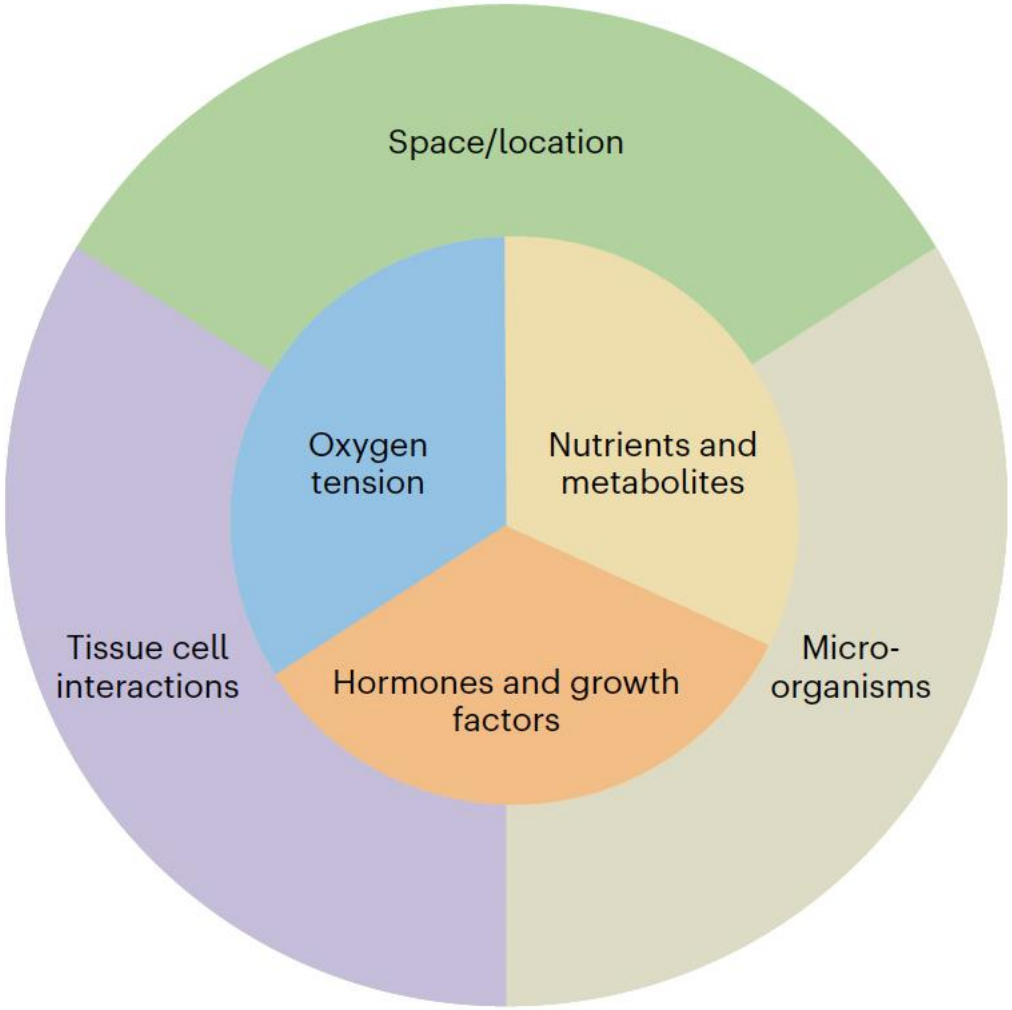
Schnell et al., 2021, Cell 184, 6281–6298

Models for antigen-presenting cell functions in inducing microbiota-specific T cell differentiation



Kedmi & Litman (2024) Immunity, 57,10, 2269

Drivers of tissue-specific signatures of immune cells



Different phenotypes: Cell types or cell states?

CD4 Effector functions: recap

- Cytokine production
- Macrophage activation
- B cell help
- DC activation/ licensing
- CTL help
- Killing

Summary

- CD4 T cells play many roles in immunity
- Different ‘types’ fulfill different functions
- In order to provide protection:
  - Proliferate
  - Differentiate/ migrate
  - Mediate effector functions
  - Generate immunological memory